



Leibniz Supercomputing Centre
of the Bavarian Academy of Sciences and Humanities

The Euro-Q-Exa Friendly User Pilot Phase: structure and tools

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The Euro-Q-Exa Friendly User Pilot Phase



Concept borrowed from HPC:

- Starting user operation with few, selected users
- No guarantee that everything runs as smooth as expected
- Users run on a system with less load
- Stress test for the system
- Tight feedback loop between LRZ consulting/support team, IQM and users



Timeline



- 2025: installation and acceptance of the system in the LRZ Compute Cube
- October 2025: MQV Annual Meeting and start of collection of user requests
- Q4/2025: preparation of the test phase, Q-Exa as preparation system

The Friendly User Pilot Phase will last until the end of May

Starting in mid-May, QCT solution architects will get in touch with you to generate a use case report (2-ish pages) that show results

Additionally, a 2-slide representation of the use case will be created

The goal is to present highlights of the Friendly User Pilot Phase at ISC26 (booth talks), with your agreement

Status and operation of the system



The system will be operated with **53** qubits

The access of the jobs to the system is mediated by a scheduler

The maximum number of execution (circuits x shots) is set to 1M

Please try to run in a way which is fair to your fellow test users

This is unfortunately a problem which has been raised by the legal team quite recently and we are currently looking for acceptable solutions

Based on the updated European Union dual-use export control regulations (effective as of late 2023/2024), quantum computers with 34 or more physical qubits are subject to strict export controls. For us it means:

- Access to Euro-Q-Exa from outside Germany, but within the EU needs an approval by the Federal Office for Economic Affairs and Export Control;
- Access to the system from non-EU locations appears legally problematic. We are looking for consulting and solutions.

The issue is the location, not much the citizenship (the export control check, done when user IDs are issued, should cover the latter)

LRZ support: the solution architects



Each research team got assigned a solution architect (actually two, for back-up) from QCT

The solution architect is your first point of contact and facilitator with the computing resources provided by LRZ

He/she will work with you on your potential issues with the computing environment and in getting the final report done

- Collection of user requests / prioritisation of software development
- Collaborative perspective, beyond technical support

IQM support



IQM brings the hardware / vendor experience in the Friendly User Pilot Phase

Shared interest of making this test a success

Available in the interaction with the users

Involved in the **collaboration assets**

The user mailing list



euro-q-exa-users@lists.lrz.de

Moderated mailing list (no messages between list members or from outside)

It includes: test users, IQM

Meant for: LRZ-to-all communications, calendar invitations

Opt-out possible, contact quantum@lrz.de for it

The Element room



Messenger and collaboration app

It includes: LRZ solution architects, test users, IQM

Meant for: all-to-all communications, short-notice announcements, (general, early) support requests

To join:

- Create a Matrix account on a Matrix server of your choice (a list of servers is at <https://servers.joinmatrix.org/>).
- Send your credentials (user ID and server), to quantum@lrz.de or to your solution architect

The Open Mic Zoom sessions

Every second Wednesday, 15:00-16:00; first instance on **March 18th**

Regular, “on-demand” meeting

It includes: LRZ solution architects, test users, IQM upon request

Meant for: in-person contact with the LRZ and IQM team; first request of support

We can propose a short “core” for announcements and other topics of general interest

Please get in touch with the LRZ team by Friday EOB of the previous week if you have intention to join

Without a core topic or a user request, the session gets cancelled.

LRZ documentation pages



<https://doku.lrz.de/euro-q-exa-friendly-user-pilot-phase-2498300715.html>

At the moment this is an empty page, which will be filled with anything suitable to facilitate your tasks.

We will make our best to maintain an informative and useful documentation, but... talk with us!

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